

LITTLE BLUE BOOK NO. 1476
Edited by E. Haldeman-Julius

What You Should Know About Your Sensations

Grace Adams

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Grace Adams

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WHAT YOU SHOULD KNOW ABOUT YOUR SENSATIONS

THE IMPORTANCE OF SENSATIONS

Since man first became interested in the processes taking place in his own mind, his dreams and emotions and thoughts, he has wondered if he could understand these relatively elusive processes as thoroughly as he could the more stable facts of the material universe. And ever since Epictetus and Aristotle began to collect the facts of mental life and formulate them into the laws which were later to become the basis of the study of psychology, philosophers and theologians and natural scientists have debated whether or not this study of the human mind constituted a real science. The champions of both sides of the question are worthy of attention.

Epictetus and Aristotle considered their psychological investigations strictly scientific. Yet as fascinating and as authoritative as their studies appeared to their contemporaries, their opinions lose value today, because none of the discipline with which they were familiar are accounted scientific by exact modern standards.

From the scientific movement to which Descartes gave such impetus in the seventeenth century until the time of Herbert Spencer, psychology was not considered a science but a mystical preoccupation not unlike theology. In Diderot's Encyclopedia of 1752 we find psychology defined as that branch of philosophy which "defines the human soul and gives an account of its activities." Bacon also classified psychology as one of the philosophies of the soul. With it he included soothsaying and witchcraft which in his day were supposed to be the two most practical applications of psychology.

Comte was the first authority to place psychology among the natural sciences instead of the philosophies. But curiously enough the one branch of psychology which he considered exact enough to be classified along with

biology was the one branch of psychology which today is thought to be completely and absolutely unscientific—phrenology.

Since the time of Spencer, who accepted psychology as an independent science following biology and preparatory to sociology, its scientific status has been fairly generally accepted. Still there have been some violent dissenters. Kant, in particular, prophesied that there could never be a science of psychology because mental phenomena could never be treated mathematically. And the old debate is still raging. John B. Watson when he adopted his behavioristic platform threw aside all of the laboriously got results of the introspectionists and functionalists because he considered them inexact, and irremediably so, when compared with the results of the biologists and chemists. Many of these same introspectionists and functionalists feel that the revolutionary and far-reaching theories of Freud and Jung and Adler are too hypothetical and too general ever to form the basis of an exact science.

Yet in one field all but the most radically behavioristic admit that traditional psychology has justified its scientific claims. This field comprises the human sensations.

THE NATURE OF SENSATIONS

Sensations are the mental ultimates of the psychological universe just as the chemical elements are the ultimates of the chemical universe. Sensations are the colors that we see—the reds and greens and purples and blacks and whites and greys; the tactual experiences that we feel—the warmths and colds and tickles and pains and aches; the tones and noises that we hear; all the vast number of odors that we smell and our few simple tastes of salt, sweet, sour and bitter. Of course these simple experiences very seldom appear isolated. They come to us as integral parts of much more complex experiences, as the taste of lemonade, the smell of a flower or the colors of a landscape. In this they parallel the elements of the other sciences which must be separated out, artificially if necessary, before they can be adequately studied.

It is a little ironical that at the very time Kant was denying that any mental phenomena could ever be treated mathematically the psychologists, themselves, were bringing the facts of sensation under mathematical laws and the natural scientists were accepting these laws as scientific, for it is to be assumed that biologists and physicists might be more jealous of the limits of an exact science than philosophers. The natural scientists, however, did more than complacently accept the psychologists' own investigations of sensations; they verified and extended them and pointed out new fields for the psychologists to explore. This was, of course, no mere altruistic gesture. Natural scientists, themselves, have a real and pertinent interest in all sensory problems and it is in their solution that we come to understand the relations between the three sciences of physics, biology and psychology, the differences among them and their interdependence on one another.

HOW SENSATIONS OCCUR

One particular physiological organ or group of organs is responsible for each of our five senses: the internal ear for sound, the retina of the eye for color, the papillae of the tongue for taste, and so on. Each of these special organs responds to a definite type of physical stimulation: the retina to vibrations of the ether, the internal ear to sound waves, the tongue to chemical solutions. Every simple psychological sensation has, then, a rather complicated causal history and it depends not only on the form of the physical stimulus but also on the peculiar make-up of its special organ.

Every physiological organ of the human body is, of course, subject to a wide range of individual differences. Some hearts are weaker than others, some livers more sluggish, and some lungs less efficient in their functions. It is only natural that the sense organs should differ also. And they do. Yet on the whole they are surprisingly uniform. All eyes react so typically to certain physical stimuli, and all noses to others, that physicists themselves call these stimuli by the names of the psychological sensations that they normally evoke. A light wave of 687 millimicrons, for example, is known as a red wave because when it impinges upon the normal retina it produces a red sensation, and a vibration of 527 is called a green wave for the same reason. A temperature of 12 degrees centigrade is marked as cold because when it stimulates the human skin it produces a cold sensation, and 45 degrees C. is hot because it produces the psychological experience of heat. Chemists are so familiar with the typical odors of certain substances that they accept these odors as invariable attributes of the substance. At times, however, our usually well behaved sense organs respond to the wrong type of stimuli and paradoxical sensations result. Neither ether vibrations nor sound waves are necessarily involved when we receive a severe blow on the head, yet such an accident can cause our ears to “ring” and our eyes to “see stars.” And a perfectly normal human being can feel a cold indistinguishable from normal cold when his skin is stimulated by a temperature of 45 degrees C. (or 113° F.). To understand this paradox it is necessary to familiarize ourselves with the anatomy and physiology of the skin.

HOW WE FEEL

The whole surface of the human skin is not equally sensitive to stimulation. In fact, a very large proportion of it is completely insensitive. But dotted throughout this insensitive area are hundreds of thousands of tiny sense organs. They are clustered together much thicker in some regions of the body than in others, and wherever they are more numerous the skin is, of course, more sensitive; yet everywhere there is a sufficient number of them to make us unaware, unless we investigate carefully, that there are any portions which are insensate. These organs are of three different kinds and each kind is responsible for a special type of sensation. The most numerous, the corpuscles of Meissner, give us our experiences of pressure, the bulbs of Kraus provide our sensations of cold and the corpuscles of Ruffini those of warmth. In a square centimeter of the skin there are approximately twenty pressure spots, thirteen cold spots and two warm spots.

When a pressure spot is touched very lightly by a small narrow object it reacts with a sensation which may be described as a tickle. The same sensation can be got by gently moving a hair on the forearm or the back of the hand for the corpuscles of Meissner are situated directly beneath and to the windward of the hair bulbs and a movement of the hairs stimulates them adequately. If the pressure is made a little stronger, the tickle becomes a sensation of contact. Both the tickle and the contact are definitely experiences of pressure. If the same pressure spot is stimulated very carefully by either a cold or a warm object, the experience will still be one of pressure. The pressure will be neither warm nor cold for the pressure spots can respond only with tactual sensations. They are as insensitive to thermal stimulations as are the many other portions of the skin where there are no temperature spots. The warm and cold spots, alone, give us all our experience of temperature.

When a cold spot is stimulated by an object whose temperature is between 12 and 15 degrees C., a cold sensation is felt. When a stimulus whose temperature is between 37 and 40 degrees C. is brought in contact with a warm spot, the typical sensation of warmth results. If, however, the

cold stimulus is applied to the warm spot, nothing at all is felt. The warm spots do not respond at all until the temperature reaches 37 degrees. Similarly the cold spots do not react to warm stimuli, warmth can be realized only from a warm spot. Yet when the temperature reaches 45 degrees C. (113° F.) the cold spots do respond, but with neither warmth nor heat. They are cold spots, not warm spots, therefore they can react in only one way—with a cold sensation. So we find that due to the physiological functioning of an anatomical sense organ, a physically hot stimulus really feels cold.

These three groups of sense organs, the pressure, cold and warm spots, are the only sense organs on the whole surface of the body. Even the pain which we feel when our skin is pricked or lacerated is not represented by a special organ. Pain is experienced whenever a nerve is stimulated directly and cutting the skin immediately exposes a nerve and pain results. The only other organs from which we receive any tactual sensations are the muscle spindles which produce aches and soreness, the spindles of the tendons from which come our sensations of strain, and the corpuscles of the joints which give us subcutaneous pressure. Yet think of the wealth of tactual experiences which we have every day, in fact, every few minutes. Not only such gross experiences as hunger and thirst and fatigue and dizziness, but simple perceptions of wetness and oiliness and smoothness and roughness and heaviness and lightness. There are no specific sense organs for any of these experiences. They are brought about by the six sense organs already mentioned and the free nerve endings of pain functioning singly or in combination, in greater or less degree.

There are not even any special sense organs of heat. Yet heat appears as a unique sensation. It is different from warmth because warmth is essentially comfortable while heat has a distinct and disagreeable sting. The characteristic sting of heat is recognized whenever we refer to pepper and other spices as hot. Psychologically, of course, the sensation of heat is unique and simple, but on its biological side it is slightly complicated. It is occasioned by the simultaneous response of a warm spot and a cold spot, or several warm and cold spots, to a stimulus of more than 45 degrees C. And it can be produced in no other way. The warm spots, of course respond normally, the cold spots paradoxically. The normal response is more vigorous than the paradoxical, consequently we do not often realize the

sensory quality of the cold, while the warmth comes into consciousness. Ordinarily, then, heat is warm. With particular stimuli and in certain regions of the body where cold spots are particularly numerous, however, trained psychologists are able to detect a heat which is cold. Also it is not an especially rare occurrence to feel cold when we plunge our hands unknowingly into very hot water.

Wetness, which seems almost as simple as heat, is produced by the simultaneous stimulation of a cold spot and a pressure spot. It is really nothing more than a cold pressure. It is true that we speak of being wet when we are in a warm bath, but it is only when the water begins to cool that we get the typical wet perception. This explains why cold objects so often feel damp when there is actually no moisture on them and why it is very difficult to determine whether cloth is dry or not when it is held close to a hot fire, also why we are so generally unconscious of the continual moisture of our own bodies.

VISUAL SENSATIONS

The number of perceptions produced by the relatively few sense organs of the skin seems quite insignificant when compared to the many different colors occasioned by the activity of the retina of the eye, for the microscopic organs in the retina bring about all of our sensations of color and of light and shade. All of the rest of the rather elaborate apparatus of the eye has to do with spacial perceptions; with form and distance and movement and size.

The normal human retina is sensitive to only some four hundred wave lengths of light. The others, like the ultra violet rays, lie outside the visible spectrum. Yet so efficient is that one small layer of the eye that it can convert these four hundred ether vibrations into approximately twelve million hues and tints and chromas. This number seems astounding because it would probably be impossible for anyone to name as many as twelve hundred separate colors. In fact, although the persons responsible for the new shades in women's stockings seem to have an unlimited color-vocabulary, most of us get along satisfactorily with an extremely limited one. Yet the fact that only a very few of these twelve million hues and shades have common names does not mean that they cannot all be detected under suitable conditions. Take the one color, black, for instance. How many vastly different shades do you group under that one word?

You can probably name immediately some ten, or maybe twenty objects which you would without hesitation describe as black. Black velvet, black satin, black cotton, black wood, coal, soot, ink, and so on. It is possible that it may never have occurred to you that these blacks are not identical. Yet a little critical attention to them will show you how very different they really are. The black paper looks decidedly grayish when placed beside the velvet. The velvet, itself, begins to pale when it is compared to the darkness of the inside of a black tube. The same variety holds for whiteness. White paper is different from white enamel and white paint. Indeed, it is somewhat difficult to find two kinds of paper whose whiteness is identical. And the

white that is got from clear sky reflected in a mirror makes the white of any object look positively dirty.

Without examining any true colors at all we have already collected a variety of shades. And this variety grows when we remember all of the various grays which are darker than the whites but lighter than the blacks. Von Kries, the German physiologist, found that there were two hundred and four different shades of gray between the blackest of black paper and the whitest of white paper. And Kulper, the psychologist, estimated that the best visible black is a thousand times darker than the best visible white.

As soon as the true colors are added to the blacks and whites and grays, the possible visual sensations begin to multiply almost limitlessly. There are literally thousands of reds: dull reds, bright reds, yellowish reds, bluish reds. And there are bluish greens, greenish blues, yellowish greens and greenish yellows. But there is a limit to these color combinations. There are no reddish greens, no greenish reds, no yellowish blues and no bluish yellows.

COMMON ABNORMALITIES OF VISION

The astonishingly wide variation of hues and tints and chromas is visible to the majority of human beings. There is a large class, however, whose visual range is much narrower. About three percent of the male population is known to be partially color blind, that is insensitive to certain visual stimuli. To such individuals blue and yellow appear as they do to persons with normal color vision; but they see both red and green as grey, greenish yellow and orange as greyish yellow, and bluish green and purple as greyish blue. Partial color blindness is hereditary and although it usually occurs only in males, it is transmitted by females. Thus a partially color blind man may have both sons and daughters who are color normal. All of the children of his sons may also have normal color vision, but the male offspring of his daughters will be partially color blind.

There are also a number of individuals who are totally color blind, who see no colors at all. Total color blindness is also hereditary and is also more prevalent among males than among females, but it is much rarer than partial color blindness and should be confused neither with that nor with true blindness. Blind persons see neither colors nor blacks nor whites nor greys. As Helmholtz put it they “see things in the same way in which we ourselves see what is behind our backs; that is to say, they do not see at all.” But totally color blind persons can see; in a dim light they can see exceptionally well. They can distinguish objects visually as well as persons with normal eyes, but for them these objects never possess any color. They always appear grey.

The two forms of color blindness seem more creditable when we realize that everyone is often either partially or totally color blind. All of us are partially color blind to objects which we see in indirect vision, or “out of the corner of the eye.” Many of us, of course, never realize this deficiency, because we usually pay very little attention to our indirect vision. If an object interests us we immediately turn our eyes toward it and focus them upon it and its colors become clear. But if you will hold an orange or a greenish blue pencil parallel to and about a foot away from your right ear

while your eyes are steadily focused directly in front of you, and move the pencil gradually until it is directly in front of your nose, you will find that at the side of your eye the orange pencil looks only yellow and the greenish blue pencil only blue. The red and green do not appear until the pencil is almost within the focus of your eyes. It is only at the very center then that the eye is color normal. It is partially color blind at the sides. And it is always totally color blind, of course, in the dark.

If we wake up suddenly on a black night or go quickly from a brightly lighted room into one that is much darker, we at first seem to be quite blind—to be incapable of distinguishing anything. Gradually, however, our eyes become accustomed to the darkness—the psychologists say that they become dark adapted, and we can make out the outlines of familiar objects, can see that parts of the room are lighter and parts darker than others. But we cannot distinguish any colors. We do not even see any clear blacks or whites. Everything appears grey. And that is the way the whole world in daylight as well as in darkness looks to the totally color blind—as a series of darker or lighter greys.

The inability to distinguish the hues of colors is not the only thing which differentiates the way we see in the dark from the way we see in daylight. We have already noted the fact that under good illumination the center of the eye is the area of clearest vision. Those objects upon which our eyes are focused are the ones which appear clearest to us. Offhand we would suspect that the same principle held in the dark. Yet it is very easy to demonstrate that it does not. Some night when you are out of doors look up at the sky and select one small isolated star and remember carefully just where it is. Then try to focus your eye upon it. You will find that this is impossible. As soon as you stare directly at it it disappears, but reappears as soon as you search for it. This process can be continued indefinitely—a disappearing and reappearing which is so rapid that the star actually seems to twinkle. Whenever you look directly at it it is gone, but as soon as you shift your eyes ever so slightly it is back again. Carefully verified experiments which elaborate this simple experience have convinced psychologists of the fact that at night all eyes are completely blind at the center where in daylight they have the clearest vision. And totally color blind eyes show this same deficiency in daylight as well as in the dark. They twitch continually when their possessors attempt to read.

So much for the psychological side of visual sensations. If we wish an explanation of them, of their variety and of their peculiarities, we must turn once more to anatomy and physiology.

HOW WE SEE IN THE LIGHT AND IN THE DARK

The retina of the eye contains a quantity of microscopic structures, some of which may be roughly described as rods, others as cones. The cones are scattered over the entire retina but are thickest at the very center. The rods have about the same distribution except that they are entirely lacking at the center. From pathological cases and from histology the function of these tiny structures has been determined.

A human eye which contains no rods has never been discovered but the eyes of fowls and pigeons show this defect, and these animals are apparently quite blind at night. All of their activities are carried on in the daylight. Hens do not even lay eggs in total darkness. Conversely the retinas of nocturnal animals, such as bats and owls and moles, are almost entirely lacking in cones but are richly supplied with rods. These and similar findings, together with the psychological differences of night and day vision, have led psychologists and biologists alike to conclude that the rods and cones, although they are very similar in structure, serve two very different functions. The cones provide us with daylight vision, apparently, while the rods, alone, are responsible for the manner in which we see in the dark. The rods are completely inactive in daylight but begin to respond to weak stimuli as soon as twilight (real or artificial) sets in. And the cones which are reacting continuously during every second that it is light, become functionally useless in the dark.

The cones alone are responsible for true colors. Many theories have been advanced to explain their functioning, but they are too long and too complicated to be reviewed here. This much can be stated, however; only three photo-chemical substances reacting to light produce all of our many different hues. The same substance reacting in different ways gives us both red and green, another both blue and yellow, and a third both black and white. When more than one substance responds to stimulation we get the intermediate colors which resemble one or more of these primary colors, for instance, purple which resembles both blue and red, and orange which is

like both red and yellow. In normal eyes all three substances are found only at the center and consequently it is only at the center that all colors are visible. In partially color blind eyes the red-green substance is absent from the center as well as the periphery and the partially color blind, therefore, can see only the colors which contain either yellow or blue. In the absence of blue or yellow stimuli they see only grey. The totally color blind eye is completely deficient in cones. The totally color blind use their rods for daylight as well as dark vision.

THE IMPORTANCE OF THE SENSE OF SMELL

We have spent so much time on visual and tactual sensations because they represent the two senses which have been investigated most thoroughly on the physical and biological side. We know not only just what these groups of sensations are psychologically but how they are physically and biologically produced. Vision has another claim to careful study. It, along with audition, is known as a higher sense and is used more by civilized man than the lower senses of touch, taste and smell. We depend upon our eyes and ears as guides throughout all of our practical life and they are the two senses which have the highest biological development. Aside from their original function of orientation, civilization has forced them to serve two still higher purposes. Our eyes and ears are responsible both for the language which we use in conversation and for our two special forms of culture—art and music. We have no highly developed gustatory or olfactory art and although both of these senses are very important in our daily lives, we never use them as conscious means of communication. Yet viewed in relation to the evolution of the race, smell is the most important of all our senses. In the extension of the olfactory organs of the reptile we find the first hint of the development of a brain. And in all animals which live on the land the sense of smell is highly specialized.

Fish have no real olfactory sensations. The piscatorial organ which corresponds to the mammalian nose is really an organ of taste. Even animals which spend the greatest part of their lives in the water—whales, dolphins and seals—have very rudimentary olfactory organs and probably no real sensations of smell. Birds, also, contrary to common opinion, have a very poor sense of smell and exceedingly efficient eyes. The power which is usually attributed to their olfactory sensitivity is actually due to their extraordinarily keen sight. It has been proven by experiments that if a decaying animal carcass is carefully concealed carrion birds will pass directly by it without pausing to investigate. In all animals, however, who habitually live upon the ground, olfactory experiences are remarkably

numerous and varied. It is easy to verify this fact by observing the behavior of insects and any four-footed mammals, especially dogs. But it is hard to believe that human beings are equally sensitive to odors. Yet they can be.

The human olfactory sense has been extensively investigated but at the present time no one has been able to state the exact number of smells which the human nostrils are capable of detecting. The chief difficulty lies in the fact that while the tactual and visual senses, for example, are physically limited, new odorous stimuli are being discovered every day. Each slightly different chemical compound produces a new odor and each new species of plant life carries its own typical smell. Already there is reasonable experimental ground for assuming that there are sensations from all of the other sense departments put together.

There are two principal reasons why we are generally unaware of this wealth of olfactory experiences. The first is that man habitually walks on two feet so that his nose, compared to that of other mammals, is very far from the ground and smells have a tendency to cling to the ground. Consequently, those smell stimuli which do reach his nostrils are very weak by the time they arrive there and are, therefore, generally ignored in favor of the more insistent visual and auditory tactual stimuli. In the second place, the most frequent and the strongest odors we experience we customarily ascribe to our sense of taste. We speak of the taste of our food when we are actually referring to its smell.

It is very easy to prove by experiment how limited is our sense of taste but we even have a very strong hint of the fact during our everyday life, or rather during that part of our everyday life when we are afflicted by a severe cold in the head. At such times we think that even the most savory meal has very little taste. Yet common sense should tell us that our tongues, which in adult life contain almost all of the sense organs of taste, are not seriously impaired by a bad cold. We should know that it is our noses, in which the olfactory organ is situated, which are really affected. It is the sense of smell which we habitually confuse with that of taste which is weakened. It is not entirely out of commission, however. There is a small passage at the back of the mouth, leading to the nose, of which we are disagreeably conscious when we choke. In swallowing, air laden with odorous particles is pushed up through this passage, the posterior nares of the nostrils are immediately stimulated and smell is experienced. This arrangement is quite evident to us

when we take medicine which has a particularly disagreeable odor. Holding the nose helps a great deal until the dose is actually swallowed. If you should continue to hold your nose and, instead of swallowing, rinse your mouth thoroughly, you would be surprised to find how very little unpleasant “taste” the medicine contained. A rather agreeable sweetness might be all that remained.

THE TRUE SENSE OF TASTE

When all odor is completely eliminated by stopping the nostrils with cotton and the surface of the tongue is stimulated directly, it has been discovered that there are only four fundamental tastes, sour, salt, sweet and bitter. A particular region of the tongue is responsible for each of these four tastes. It is very easy to find these specialized parts for, while the functional organ of the nose is so securely hidden that it is almost impossible to stimulate it directly, the gustatory organs are visible to the naked eye. They are contained in the tiny bright red spots, or papillae, that dot the duller surface of the tongue everywhere except in the center. Although any single papilla is apt to contain several kinds of taste buds and may, therefore, be sensitive to more than one taste, and although the buds which respond to salt are fairly equally distributed over the sensitive area of the tongue, those which react to sweet, sour and bitter are grouped on different regions. Language recognizes this distribution.

We describe as a sweet expression one in which the lips are drawn toward the tip of the tongue and the tip, itself, is flattened out. The taste buds which respond to sweet stimuli are found clustered at the tip of the tongue. In a sour expression the lips are drawn back from the tip of the tongue and down toward the sides. And the sour taste buds are thickest on the sides of the tongue. When a face wears a bitter expression the tongue itself is pulled back until it is in contact with the roof of the mouth. And the bitter taste buds are found almost exclusively at the rear of the tongue.

Testing these three facial expressions before the mirror will afford you a little innocent amusement, but remembering what they represent will help you to get the full effect of desirable tastes and to avoid the disagreeable qualities of others. A bitter tablet, for example, should not be placed at the rear of the tongue where you would get the full benefit of its bitterness, but rather in the center which is relatively insensitive to any taste.

The manner in which children lick a stick of candy shows that they realize unconsciously how they can make the most of its sweetness. The taste buds also explain another habit of children which is usually frowned

upon by their elders—that of taking large mouthfuls. Children have many more taste buds than adults. They are distributed not only over the whole surface of the tongue but on the insides of the cheeks as well. Consequently when children stuff their mouths until their jaws bulge, they actually get a great many more gustatory sensations than when they have only a dainty morsel on their tongues. If the taste buds in the cheeks did not atrophy early in life, there would probably be nothing ill mannered about large mouthfuls. As it is, large mouthfuls are quite useless to adults. The food has to be directly on their tongues before they can taste it. So, forgetting the days when bulging cheeks were so delightful, adults become intolerant of a habit which was once both pleasant and useful.

THE COMPLEXITY OF OUR PERCEPTIONS

Because it is possible to isolate particular sensations and study them in detail, as we have attempted to do here, it must not be assumed that we customarily experience them in so simple a manner. Even at birth the human nervous system is a tremendously complicated affair. There are cross connections among the senses. From the very first most of our sensations seem to us completely bound up with other processes. A pure sensation is a very rare occurrence. A color is nearly always the color of something—of cloth or leather or paper or liquid. All of the objects which have colors have textures also. They look rough or smooth or glossy or dull. And the color and texture seem inalterably mixed together. It is only when we look at a rainbow or the rays of a spectroscope that we see a color which does not appear to be an integral part of some object.

Tones seem to belong less to the objects which produce them—to exist by themselves to a greater degree. Yet it is much more difficult to produce a simple tone than a pure color. In this case the trouble lies not so much with the nervous system as in the physical stimulus itself. There are not as many audible tones as there are discernible colors and odors, but there are some eleven thousand of them. The tone got from a musical instrument is not, however, a simple tone or a pure sensation. It is a combination of many tones, some of which are more intense than others. These extra tones or partials are produced by the musical instrument itself and they cannot be eliminated because a musical instrument is essentially a complicated vibrating body. With each vibration it produces over-tones as well as a principal tone. It is the different intensities of the partial tones which give timbre to instruments. It is by their characteristic timbres that we are able to distinguish a G struck on a piano from the G of an organ or a violin or a saxophone.

Because it is so difficult to produce simple tones, psychologists have been unable to discover a great deal about our sense of audition. The physical side of sound has been adequately investigated, but we have to

resort to musical terminology when we try to describe the sensations which are produced.

We have already discovered, then, in the case of the three lower senses, our perceptions appear so unified that we seldom realize that they are complex. Wetness seems to be a unique experience until we recognize that it is composed of cold and pressure. So completely blended are our tastes and smells that a severe cold in the head is required for their conscious separation. And psychologists had to wait for years, until a soldier had a strange cortical injury, before they learned how complicated a process is tactual localization.

LOCALIZATION ON THE SKIN

Localizing a spot on the surface of the skin seems to be the very simplest of matters. An itch occurs on the hand and automatically we scratch it. Sometimes, it is true, the itch comes in the small of the back and we cannot quite reach it. But we can at least point out its approximate position to some other person. Both the scratching and the pointing seem to be as simple as reflex actions. This apparent simplicity led the psychologist Loetze to believe that every single spot on the body had a special attribute which he called a "local sign." According to his doctrine we know exactly where we have been touched the instant that we are aware of the touch itself. Henri, another psychologist, discovered, however, that if he was stimulated on the skin when his eyes were closed and tried to locate the exact spot before he opened his eyes again, he made large errors of localization and the errors were as great on the finger as on the forearm. This discovery cast grave doubt on the doctrine of the local sign.

During the late war two psychologists, Gelb and Goldstein, were attached to a German army hospital. One day a patient was brought in, who had had both of his occipital lobes destroyed, but the rest of his nervous system was still intact. This meant that he had absolutely no visual cognitions but the rest of his mental functions were unimpaired. Gelb and Goldstein immediately began to experiment with him and they got some rather startling results.

If the patient was lying quietly with his eyes shut and was touched on some part of his body, perhaps his left foot, he knew that he had been touched but he had no idea of the place. He did not know whether it was on the leg or the forehead or the chest or the hand. It made no difference how intense the stimulus was or how long it was continued. As long as he remained motionless he could not tell where it was. If, however, he was allowed to move his body as much as he wished, he could localize the place touched almost exactly but he did it in a very strange way. He at first wriggled about so that he set his entire body in motion. Then gradually he would move less and less. If the stimulus was applied to his right index

finger, he would, after a few squirms of his whole body, move only his trunk, then only his right arm, then just his forearm, his right hand, his fingers, and finally only the stimulated finger. Then he could say just where the pressure was felt.

If he was touched in two places at once, as long as he was quiet, he would feel only one. If the two stimuli happened to be very close together, for instance both on the same finger, even by wriggling he could not discover that they were two. But if they were rather widely separated he could by the proper movements locate first one and then the other.

When he was allowed to touch the place stimulated, his method was the same as when he wriggled his body. At first he would run his finger aimlessly over his skin, gradually bringing it closer and closer to the stimulated area until the exact spot was hit upon.

These results led Gelb and Goldstein to the conclusion, which was foreshadowed by Henri, that immediate localization is always a function of vision. And they agreed with the findings of some rather different experiments which had been carried on at Cornell University several years before. The subjects of these experiments had to be real scientific martyrs. They were required to swallow a rubber tube in which there was an electric coil. The tube reached from the mouth to the bottom of the stomach and at intervals there were openings in it through which the experimenter could send an electric shock. All of the subjects had the greatest difficulty in locating the shocks; they could feel them clearly enough but they could not discover in which part of their digestive tracts they were. A stimulation might be at the extreme end of the tube and the subject would report that it was in his thorax. A shock at the esophagus would be referred to the bottom of the stomach, and so on.

If localization is primarily a matter of vision, these errors are easily accounted for. No one has a very complete mental picture of the inside of his own digestive tract. When we localize sensations there we must do so by reference to the more familiar external body. But since the sensations are felt internally, not on the outside, it is only natural that the reference should often be erroneous. Doctors recognize this. They accept a patient's statement that he has a pain but they are not so quick to take his word for its exact location. Instead they investigate. And when they thump about first at

one place then at another, they are using the same method of localization which Gelb and Goldstein's soldier employed when he ran his finger over his body. Fortunately, this method, though much slower than visual localization, is usually successful.

Transcriber's Note:

Inconsistencies in hyphenation have been standardized.

Minor punctuation errors have been changed without notice.

Spelling was retained as in the original except for the following changes:

Page [21](#): “spent so much time is” to “spent so much time on”

Page [27](#): “most of our sensations some” to “most of our sensations seem”

Page [27](#): “a pure sensation. It it” to “a pure sensation. It is”

***** END OF THE PROJECT GUTENBERG EBOOK WHAT YOU
SHOULD KNOW ABOUT YOUR SENSATIONS *****

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